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Electronic Devices Having Multilayer Millimeter Wave Antennas

Abstract

An electronic device may have a phased antenna array. An antenna in the array may include a rectangular patch element with diagonal axes. The antenna may have first and second antenna feeds coupled to the patch element along the diagonal axes. The antenna may be rotated at a forty-five degree angle relative to other antennas in the array. The antenna may have one or two layers of parasitic elements overlapping the patch element. For example, the antenna may have a layer of coplanar parasitic patches separated by a gap. The antenna may also have an additional parasitic patch that is located farther from the patch element than the layer of coplanar parasitic patches. The additional parasitic patch may overlap the patch element and the gap in the coplanar parasitic patches. The antenna may exhibit a relatively small footprint and minimal mutual coupling with other antennas in the array.

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Background/Summary

[0001] This application is a continuation of U.S. patent application Ser. No. 18/521,521, filed Nov. 28, 2023, which is a continuation of U.S. patent application Ser. No. 17/751,495, filed May 23, 2022, now U.S. Pat. No. 11,894,608, which is a continuation of U.S. patent application Ser. No. 17/028,864, filed Sep. 22, 2020, now U.S. Pat. No. 11,349,204, each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] This relates generally to electronic devices and, more particularly, to electronic devices with wireless communications circuitry.

[0003] Electronic devices often include wireless communications circuitry. For example, cellular telephones, computers, and other devices often contain antennas and wireless transceivers for supporting wireless communications.

[0004] It may be desirable to support wireless communications in millimeter wave and centimeter wave communications bands. Millimeter wave communications, which are sometimes referred to as extremely high frequency (EHF) communications, and centimeter wave communications involve communications at frequencies of about 10-300 GHz. Operation at these frequencies may support high bandwidths, but may raise significant challenges. For example, millimeter wave communications signals generated by antennas can be characterized by substantial attenuation and/or distortion during signal propagation through various mediums. In addition, if care is not taken, the antennas can be susceptible to undesirable mutual coupling.

[0005] It would therefore be desirable to be able to provide electronic devices with improved wireless communications circuitry such as communications circuitry that supports millimeter and centimeter wave communications.

SUMMARY

[0006] An electronic device may be provided with wireless circuitry. The wireless circuitry may include a phased antenna array. The phased antenna array may convey radio-frequency signals in a signal beam at a frequency greater than 10 GHz.

[0007] An antenna in the phased antenna array may include a rectangular patch element. The rectangular patch element may have first and second diagonal axes. The antenna may have a first positive antenna feed terminal coupled to the rectangular patch element along the first diagonal axis. The antenna may have a second positive antenna feed terminal coupled to the rectangular patch element along the second diagonal axis. The antenna may be rotated at a forty-five degree angle with respect to adjacent antennas in the phased antenna array.

[0008] The antenna may have parasitic elements overlapping the patch element. For example, the antenna may have five parasitics formed in a single layer overlapping the patch element. Gaps may separate each of the parasitics from each other. As another example, the antenna may have a layer of coplanar parasitic patches overlapping the patch element. The parasitic patches in this layer may

be separated by a gap. The antenna may also have an additional parasitic patch that is located farther from the patch element than the layer of coplanar parasitic patches. The additional parasitic patch may overlap the patch element and the gap in the layer of coplanar parasitic patches. When configured in this way, the antenna may exhibit a relatively small footprint and minimal mutual coupling with other antennas in the array.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a front perspective view of an illustrative electronic device with wireless circuitry in accordance with some embodiments.

[0010] FIG. 2 is a rear perspective view of an illustrative electronic device with wireless circuitry in accordance with some embodiments.

[0011] FIG. 3 is a schematic diagram of an illustrative electronic device with wireless circuitry in accordance with some embodiments.

[0012] FIG. 4 is a diagram of an illustrative phased antenna array that forms a radio-frequency signal beam at different beam pointing angles in accordance with some embodiments.

[0013] FIG. 5 is a diagram of illustrative wireless circuitry in accordance with some embodiments.

[0014] FIG. 6 is a top view of an illustrative antenna having diagonally-oriented feed terminals in accordance with some embodiments.

[0015] FIG. 7 is a cross-sectional side view of an illustrative antenna having diagonally-oriented feed terminals in accordance with some embodiments.

[0016] FIG. 8 is a plot of antenna performance (mutual coupling) as a function of frequency for an illustrative antenna having diagonally-oriented feed terminals in accordance with some embodiments.

[0017] FIG. 9 is a top view of an illustrative antenna having multi-layer parasitic elements in accordance with some embodiments.

[0018] FIG. 10 is a cross-sectional side view of an illustrative antenna having multi-layer parasitic elements in accordance with some embodiments.

[0019] FIG. 11 is a plot of antenna performance (return loss) as a function of frequency for illustrative antennas in accordance with some embodiments.

DETAILED DESCRIPTION

[0020] An electronic device such as electronic device 10 of FIG. 1 may contain wireless circuitry. The wireless circuitry may include one or more antennas. The antennas may include phased antenna arrays that are used for performing wireless communications and/or spatial ranging operations using millimeter and centimeter wave signals. Millimeter wave signals, which are sometimes referred to as extremely high frequency (EHF) signals, propagate at frequencies above about 30 GHz (e.g., at 60 GHz or other frequencies between about 30 GHz and 300 GHz).

Centimeter wave signals propagate at frequencies between about 10 GHz and 30 GHz. If desired, device 10 may also contain antennas for handling satellite navigation system signals, cellular telephone signals, local wireless area network signals, near-field communications, light-based wireless communications, or other wireless communications.

[0021] Electronic device 10 may be a computing device such as a laptop computer, a computer monitor containing an embedded computer, a tablet computer, a cellular telephone, a media player, or other handheld or portable electronic device, a smaller device such as a wristwatch device, a pendant device, a headphone or earpiece device, a virtual or augmented reality headset device, a device embedded in eyeglasses or other equipment worn on a user's head, or other wearable or miniature device, a television, a computer display that does not contain an embedded computer, a gaming device, a navigation device, an embedded system such as a system in which electronic

equipment with a display is mounted in a kiosk or automobile, a wireless access point or base station, a desktop computer, a portable speaker, a keyboard, a gaming controller, a gaming system, a computer mouse, a mousepad, a trackpad or touchpad, equipment that implements the functionality of two or more of these devices, or other electronic equipment. In the illustrative configuration of FIG. 1, device **10** is a portable device such as a cellular telephone, media player, tablet computer, portable speaker, or other portable computing device. Other configurations may be used for device **10** if desired. The example of FIG. 1 is merely illustrative.

[0022] As shown in FIG. 1, device **10** may include a display such as display **8**. Display **8** may be mounted in a housing such as housing **12**. Housing **12**, which may sometimes be referred to as an enclosure or case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, etc.), other suitable materials, or a combination of any two or more of these materials. Housing **12** may be formed using a unibody configuration in which some or all of housing **12** is machined or molded as a single structure or may be formed using multiple structures (e.g., an internal frame structure, one or more structures that form exterior housing surfaces, etc.).

[0023] Display **8** may be a touch screen display that incorporates a layer of conductive capacitive touch sensor electrodes or other touch sensor components (e.g., resistive touch sensor components, acoustic touch sensor components, force-based touch sensor components, light-based touch sensor components, etc.) or may be a display that is not touch-sensitive. Capacitive touch sensor electrodes may be formed from an array of indium tin oxide pads or other transparent conductive structures.

[0024] Display **8** may include an array of display pixels formed from liquid crystal display (LCD) components, an array of electrophoretic display pixels, an array of plasma display pixels, an array of organic light-emitting diode display pixels, an array of electrowetting display pixels, or display pixels based on other display technologies.

[0025] Display **8** may be protected using a display cover layer such as a layer of transparent glass, clear plastic, sapphire, or other transparent dielectrics. Openings may be formed in the display cover layer. For example, openings may be formed in the display cover layer to accommodate one or more buttons, sensor circuitry such as a fingerprint sensor or light sensor, ports such as a speaker port or microphone port, etc. Openings may be formed in housing **12** to form communications ports (e.g., an audio jack port, a digital data port, charging port, etc.). Openings in housing **12** may also be formed for audio components such as a speaker and/or a microphone.

[0026] Antennas may be mounted in housing **12**. If desired, some of the antennas (e.g., antenna arrays that implement beam steering, etc.) may be mounted under an inactive border region of display **8** (see, e.g., illustrative antenna locations **6** of FIG. 1). Display **8** may contain an active area with an array of pixels (e.g., a central rectangular portion). Inactive areas of display **8** are free of pixels and may form borders for the active area. If desired, antennas may also operate through dielectric-filled openings in the rear of housing **12** or elsewhere in device **10**.

[0027] To avoid disrupting communications when an external object such as a human hand or other body part of a user blocks one or more antennas, antennas may be mounted at multiple locations in housing **12**. Sensor data such as proximity sensor data, real-time antenna impedance measurements, signal quality measurements such as received signal strength information, and other data may be used in determining when one or more antennas is being adversely affected due to the orientation of housing **12**, blockage by a user's hand or other external object, or other environmental factors. Device **10** can then switch one or more replacement antennas into use in place of the antennas that are being adversely affected.

[0028] Antennas may be mounted at the corners of housing **12** (e.g., in corner locations **6** of FIG. 1 and/or in corner locations on the rear of housing **12**), along the peripheral edges of housing **12**, on the rear of housing **12**, under the display cover glass or other dielectric display cover layer that is used in covering and protecting display **8** on the front of device **10**, over a dielectric window on a rear face of housing **12** or the edge of housing **12**, over a dielectric cover layer such as a dielectric

rear housing wall that covers some or all of the rear face of device **10**, or elsewhere in device **10**. [0029] FIG. **2** is a rear perspective view of electronic device **10** showing illustrative locations **6** on the rear and sides of housing **12** in which antennas (e.g., single antennas and/or phased antenna arrays) may be mounted in device **10**. The antennas may be mounted at the corners of device **10**, along the edges of housing **12** such as edges formed by sidewalls **12E**, on upper and lower portions of rear housing wall **12R**, in the center of rear housing wall **12R** (e.g., under a dielectric window structure or other antenna window in the center of rear housing wall **12R**), at the corners of rear housing wall **12R** (e.g., on the upper left corner, upper right corner, lower left corner, and lower right corner of the rear of housing **12** and device **10**), etc.

[0030] In configurations in which housing **12** is formed entirely or nearly entirely from a dielectric (e.g., plastic, glass, sapphire, ceramic, fabric, etc.), the antennas may transmit and receive antenna signals through any suitable portion of the dielectric. In configurations in which housing **12** is formed from a conductive material such as metal, regions of the housing such as slots or other openings in the metal may be filled with plastic or other dielectrics. The antennas may be mounted in alignment with the dielectric in the openings. These openings, which may sometimes be referred to as dielectric antenna windows, dielectric gaps, dielectric-filled openings, dielectric-filled slots, elongated dielectric opening regions, etc., may allow antenna signals to be transmitted to external wireless equipment from the antennas mounted within the interior of device **10** and may allow internal antennas to receive antenna signals from external wireless equipment. In another suitable arrangement, the antennas may be mounted on the exterior of conductive portions of housing **12**.

[0031] FIGS. **1** and **2** are merely illustrative. In general, housing **12** may have any desired shape (e.g., a rectangular shape, a cylindrical shape, a spherical shape, combinations of these, etc.). Display **8** of FIG. **1** may be omitted if desired. Antennas may be located within housing **12**, on housing **12**, and/or external to housing **12**.

[0032] A schematic diagram of illustrative components that may be used in device **10** is shown in FIG. **3**. As shown in FIG. **3**, device **10** may include control circuitry **14**. Control circuitry **14** may include storage such as storage circuitry **20**. Storage circuitry **20** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc.

[0033] Control circuitry **14** may include processing circuitry such as processing circuitry **22**. Processing circuitry **22** may be used to control the operation of device **10**. Processing circuitry **22** may include on one or more microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, central processing units (CPUs), etc. Control circuitry **14** may be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on storage circuitry **20** (e.g., storage circuitry **20** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on storage circuitry **20** may be executed by processing circuitry **22**.

[0034] Control circuitry **14** may be used to run software on device **10** such as internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **14** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **14** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols—sometimes referred to as WiFi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other WPAN protocols, IEEE 802.11ad protocols, cellular telephone protocols, MIMO protocols, antenna diversity protocols, satellite navigation system protocols,

antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), etc. Each communication protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

[0035] Device **10** may include input-output circuitry **16**. Input-output circuitry **16** may include input-output devices **18**. Input-output devices **18** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **18** may include user interface devices, data port devices, sensors, and other input-output components. For example, input-output devices may include touch screens, displays without touch sensor capabilities, buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, speakers, status indicators, light sources, audio jacks and other audio port components, digital data port devices, light sensors, gyroscopes, accelerometers or other components that can detect motion and device orientation relative to the Earth, capacitance sensors, proximity sensors (e.g., a capacitive proximity sensor and/or an infrared proximity sensor), magnetic sensors, and other sensors and input-output components.

[0036] Input-output circuitry **16** may include wireless circuitry such as wireless circuitry **24** for wirelessly conveying radio-frequency signals. While control circuitry **14** is shown separately from wireless circuitry **24** in the example of FIG. **3** for the sake of clarity, wireless circuitry **24** may include processing circuitry that forms a part of processing circuitry **22** and/or storage circuitry that forms a part of storage circuitry **20** of control circuitry **14** (e.g., portions of control circuitry **14** may be implemented on wireless circuitry **24**). As an example, control circuitry **14** may include baseband processor circuitry or other control components that form a part of wireless circuitry **24**.

[0037] Wireless circuitry **24** may include millimeter and centimeter wave transceiver circuitry such as millimeter/centimeter wave transceiver circuitry **28**. Millimeter/centimeter wave transceiver circuitry **28** may support communications at frequencies between about 10 GHz and 300 GHz. For example, millimeter/centimeter wave transceiver circuitry **28** may support communications in Extremely High Frequency (EHF) or millimeter wave communications bands between about 30 GHz and 300 GHz and/or in centimeter wave communications bands between about 10 GHz and 30 GHz (sometimes referred to as Super High Frequency (SHF) bands). As examples, millimeter/centimeter wave transceiver circuitry **28** may support communications in an IEEE K communications band between about 18 GHz and 27 GHz, a Ka communications band between about 26.5 GHz and 40 GHz, a Ku communications band between about 12 GHz and 18 GHz, a V communications band between about 40 GHz and 75 GHz, a W communications band between about 75 GHz and 110 GHz, or any other desired frequency band between approximately 10 GHz and 300 GHz. If desired, millimeter/centimeter wave transceiver circuitry **28** may support IEEE 802.11ad communications at 60 GHz and/or 5th generation mobile networks or 5th generation wireless systems (5G) New Radio (NR) Frequency Range 2 (FR2) communications bands between about 24 GHz and 90 GHz. Millimeter/centimeter wave transceiver circuitry **28** may be formed from one or more integrated circuits (e.g., multiple integrated circuits mounted on a common printed circuit in a system-in-package device, one or more integrated circuits mounted on different substrates, etc.).

[0038] Millimeter/centimeter wave transceiver circuitry **28** (sometimes referred to herein simply as transceiver circuitry **28** or millimeter/centimeter wave circuitry **28**) may perform spatial ranging operations using radio-frequency signals at millimeter and/or centimeter wave frequencies that are transmitted and received by millimeter/centimeter wave transceiver circuitry **28**. The received signals may be a version of the transmitted signals that have been reflected off of external objects and back towards device **10**. Control circuitry **14** may process the transmitted and received signals to detect or estimate a range between device **10** and one or more external objects in the surroundings of device **10** (e.g., objects external to device **10** such as the body of a user or other

persons, other devices, animals, furniture, walls, or other objects or obstacles in the vicinity of device **10**). If desired, control circuitry **14** may also process the transmitted and received signals to identify a two or three-dimensional spatial location of the external objects relative to device **10**. [0039] Spatial ranging operations performed by millimeter/centimeter wave transceiver circuitry **28** are unidirectional. If desired, millimeter/centimeter wave transceiver circuitry **28** may also perform bidirectional communications with external wireless equipment such as external wireless equipment **10'** (e.g., over bi-directional millimeter/centimeter wave wireless communications link **31**). External wireless equipment **10'** may include other electronic devices such as electronic device **10**, a wireless base station, wireless access point, a wireless accessory, or any other desired equipment that transmits and receives millimeter/centimeter wave signals. Bidirectional communications involve both the transmission of wireless data by millimeter/centimeter wave transceiver circuitry **28** and the reception of wireless data that has been transmitted by external wireless equipment **10'**. The wireless data may, for example, include data that has been encoded into corresponding data packets such as wireless data associated with a telephone call, streaming media content, internet browsing, wireless data associated with software applications running on device **10**, email messages, etc.

[0040] If desired, wireless circuitry **24** may include transceiver circuitry for handling communications at frequencies below 10 GHz such as non-millimeter/centimeter wave transceiver circuitry **26**. For example, non-millimeter/centimeter wave transceiver circuitry **26** may handle wireless local area network (WLAN) communications bands such as the 2.4 GHz and 5 GHz Wi-Fi® (IEEE 802.11) bands, wireless personal area network (WPAN) communications bands such as the 2.4 GHz Bluetooth® communications band, cellular telephone communications bands such as a cellular low band (LB) (e.g., 600 to 960 MHz), a cellular low-midband (LMB) (e.g., 1400 to 1550 MHz), a cellular midband (MB) (e.g., from 1700 to 2200 MHz), a cellular high band (HB) (e.g., from 2300 to 2700 MHz), a cellular ultra-high band (UHB) (e.g., from 3300 to 5000 MHz, or other cellular communications bands between about 600 MHz and about 5000 MHz (e.g., 3G bands, 4G LTE bands, 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, etc.), a near-field communications (NFC) band (e.g., at 13.56 MHz), satellite navigations bands (e.g., an L1 global positioning system (GPS) band at 1575 MHz, an L5 GPS band at 1176 MHz, a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, etc.), ultra-wideband (UWB) communications band(s) supported by the IEEE 802.15.4 protocol and/or other UWB communications protocols (e.g., a first UWB communications band at 6.5 GHz and/or a second UWB communications band at 8.0 GHz), and/or any other desired communications bands. The communications bands handled by the radio-frequency transceiver circuitry may sometimes be referred to herein as frequency bands or simply as “bands,” and may span corresponding ranges of frequencies. Non-millimeter/centimeter wave transceiver circuitry **26** and millimeter/centimeter wave transceiver circuitry **28** may each include one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive radio-frequency components, switching circuitry, transmission line structures, and other circuitry for handling radio-frequency signals.

[0041] In general, the transceiver circuitry in wireless circuitry **24** may cover (handle) any desired frequency bands of interest. As shown in FIG. 3, wireless circuitry **24** may include antennas **30**. The transceiver circuitry may convey radio-frequency signals using one or more antennas **30** (e.g., antennas **30** may convey the radio-frequency signals for the transceiver circuitry). The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external wireless communications equipment). Antennas **30** may transmit the radio-frequency signals by radiating the radio-frequency signals into free space (or to freespace through intervening device structures such as a dielectric cover layer). Antennas **30** may additionally or alternatively receive the radio-frequency signals from free space (e.g., through intervening devices structures such as a dielectric cover layer). The transmission and reception of

radio-frequency signals by antennas **30** each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna by the radio-frequency signals within the frequency band(s) of operation of the antenna.

[0042] In satellite navigation system links, cellular telephone links, and other long-range links, radio-frequency signals are typically used to convey data over thousands of feet or miles. In Wi-Fi® and Bluetooth® links at 2.4 and 5 GHz and other short-range wireless links, radio-frequency signals are typically used to convey data over tens or hundreds of feet. Millimeter/centimeter wave transceiver circuitry **28** may convey radio-frequency signals over short distances that travel over a line-of-sight path. To enhance signal reception for millimeter and centimeter wave communications, phased antenna arrays and beam forming (steering) techniques may be used (e.g., schemes in which antenna signal phase and/or magnitude for each antenna in an array are adjusted to perform beam steering). Antenna diversity schemes may also be used to ensure that the antennas that have become blocked or that are otherwise degraded due to the operating environment of device **10** can be switched out of use and higher-performing antennas used in their place.

[0043] Antennas **30** in wireless circuitry **24** may be formed using any suitable antenna types. For example, antennas **30** may include antennas with resonating elements that are formed from stacked patch antenna structures, loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, monopole antenna structures, dipole antenna structures, helical antenna structures, Yagi (Yagi-Uda) antenna structures, hybrids of these designs, etc. If desired, one or more of antennas **30** may be cavity-backed antennas. Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a non-millimeter/centimeter wave wireless link for non-millimeter/centimeter wave transceiver circuitry **26** and another type of antenna may be used in conveying radio-frequency signals at millimeter and/or centimeter wave frequencies for millimeter/centimeter wave transceiver circuitry **28**. Antennas **30** that are used to convey radio-frequency signals at millimeter and centimeter wave frequencies may be arranged in one or more phased antenna arrays. In one suitable arrangement that is described herein as an example, the antennas **30** that are arranged in a corresponding phased antenna array may be stacked patch antennas having patch antenna resonating elements that overlap and are vertically stacked with respect to one or more parasitic patch elements.

[0044] FIG. **4** is a diagram showing how antennas **30** for handling radio-frequency signals at millimeter and centimeter wave frequencies may be formed in a phased antenna array. As shown in FIG. **4**, phased antenna array **36** (sometimes referred to herein as array **36**, antenna array **36**, or array **36** of antennas **30**) may be coupled to radio-frequency transmission line paths **32**. For example, a first antenna **30-1** in phased antenna array **36** may be coupled to a first radio-frequency transmission line path **32-1**, a second antenna **30-2** in phased antenna array **36** may be coupled to a second radio-frequency transmission line path **32-2**, an Mth antenna **30-M** in phased antenna array **36** may be coupled to an Mth radio-frequency transmission line path **32-M**, etc. While antennas **30** are described herein as forming a phased antenna array, the antennas **30** in phased antenna array **36** may sometimes also be referred to as collectively forming a single phased array antenna (e.g., where each antenna **30** in the phased array antenna forms an antenna element of the phased array antenna).

[0045] Radio-frequency transmission line paths **32** may each be coupled to millimeter/centimeter wave transceiver circuitry **28** of FIG. **3**. Each radio-frequency transmission line path **32** may include one or more radio-frequency transmission lines, a positive signal conductor, and a ground signal conductor. The positive signal conductor may be coupled to a positive antenna feed terminal on an antenna resonating element of the corresponding antenna **30**. The ground signal conductor may be coupled to a ground antenna feed terminal on an antenna ground for the corresponding antenna **30**.

[0046] Radio-frequency transmission line paths **32** may include stripline transmission lines

(sometimes referred to herein simply as striplines), coaxial cables, coaxial probes realized by metalized vias, microstrip transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, waveguide structures, conductive vias, combinations of these, etc. Multiple types of transmission lines may be used to couple the millimeter/centimeter wave transceiver circuitry to phased antenna array **36**. Filter circuitry, switching circuitry, impedance matching circuitry, phase shifter circuitry, amplifier circuitry, and/or other circuitry may be interposed on radio-frequency transmission line path **32**, if desired.

[0047] Radio-frequency transmission lines in device **10** may be integrated into ceramic substrates, rigid printed circuit boards, and/or flexible printed circuits. In one suitable arrangement, radio-frequency transmission lines in device **10** may be integrated within multilayer laminated structures (e.g., layers of a conductive material such as copper and a dielectric material such as a resin that are laminated together without intervening adhesive) that may be folded or bent in multiple dimensions (e.g., two or three dimensions) and that maintain a bent or folded shape after bending (e.g., the multilayer laminated structures may be folded into a particular three-dimensional shape to route around other device components and may be rigid enough to hold its shape after folding without being held in place by stiffeners or other structures). All of the multiple layers of the laminated structures may be batch laminated together (e.g., in a single pressing process) without adhesive (e.g., as opposed to performing multiple pressing processes to laminate multiple layers together with adhesive).

[0048] Antennas **30** in phased antenna array **36** may be arranged in any desired number of rows and columns or in any other desired pattern (e.g., the antennas need not be arranged in a grid pattern having rows and columns). During signal transmission operations, radio-frequency transmission line paths **32** may be used to supply signals (e.g., radio-frequency signals such as millimeter wave and/or centimeter wave signals) from millimeter/centimeter wave transceiver circuitry **28** (FIG. 3) to phased antenna array **36** for wireless transmission. During signal reception operations, radio-frequency transmission line paths **32** may be used to convey signals received at phased antenna array **36** (e.g., from external wireless equipment **10'** of FIG. 3) to millimeter/centimeter wave transceiver circuitry **28** (FIG. 3).

[0049] The use of multiple antennas **30** in phased antenna array **36** allows radio-frequency beam forming arrangements (sometimes referred to herein as radio-frequency beam steering arrangements) to be implemented by controlling the relative phases and magnitudes (amplitudes) of the radio-frequency signals conveyed by the antennas. In the example of FIG. 4, the antennas **30** in phased antenna array **36** each have a corresponding radio-frequency phase and magnitude controller **33** (e.g., a first phase and magnitude controller **33-1** interposed on radio-frequency transmission line path **32-1** may control phase and magnitude for radio-frequency signals handled by antenna **30-1**, a second phase and magnitude controller **33-2** interposed on radio-frequency transmission line path **32-2** may control phase and magnitude for radio-frequency signals handled by antenna **30-2**, an Mth phase and magnitude controller **33-M** interposed on radio-frequency transmission line path **32-M** may control phase and magnitude for radio-frequency signals handled by antenna **30-M**, etc.).

[0050] Phase and magnitude controllers **33** may each include circuitry for adjusting the phase of the radio-frequency signals on radio-frequency transmission line paths **32** (e.g., phase shifter circuits) and/or circuitry for adjusting the magnitude of the radio-frequency signals on radio-frequency transmission line paths **32** (e.g., power amplifier and/or low noise amplifier circuits). Phase and magnitude controllers **33** may sometimes be referred to collectively herein as beam steering or beam forming circuitry (e.g., beam steering circuitry that steers the beam of radio-frequency signals transmitted and/or received by phased antenna array **36**).

[0051] Phase and magnitude controllers **33** may adjust the relative phases and/or magnitudes of the transmitted signals that are provided to each of the antennas in phased antenna array **36** and may adjust the relative phases and/or magnitudes of the received signals that are received by phased

antenna array **36**. Phase and magnitude controllers **33** may, if desired, include phase detection circuitry for detecting the phases of the received signals that are received by phased antenna array **36**. The term “beam,” “signal beam,” “radio-frequency beam,” or “radio-frequency signal beam” may be used herein to collectively refer to wireless signals that are transmitted and received by phased antenna array **36** in a particular direction. The signal beam may exhibit a peak gain that is oriented in a particular beam pointing direction at a corresponding beam pointing angle (e.g., based on constructive and destructive interference from the combination of signals from each antenna in the phased antenna array). The term “transmit beam” may sometimes be used herein to refer to radio-frequency signals that are transmitted in a particular direction whereas the term “receive beam” may sometimes be used herein to refer to radio-frequency signals that are received from a particular direction.

[0052] If, for example, phase and magnitude controllers **33** are adjusted to produce a first set of phases and/or magnitudes for transmitted radio-frequency signals, the transmitted signals will form a transmit beam as shown by beam **B1** of FIG. **4** that is oriented in the direction of point A. If, however, phase and magnitude controllers **33** are adjusted to produce a second set of phases and/or magnitudes for the transmitted signals, the transmitted signals will form a transmit beam as shown by beam **B2** that is oriented in the direction of point B. Similarly, if phase and magnitude controllers **33** are adjusted to produce the first set of phases and/or magnitudes, radio-frequency signals (e.g., radio-frequency signals in a receive beam) may be received from the direction of point A, as shown by beam **B1**. If phase and magnitude controllers **33** are adjusted to produce the second set of phases and/or magnitudes, radio-frequency signals may be received from the direction of point B, as shown by beam **B2**.

[0053] Each phase and magnitude controller **33** may be controlled to produce a desired phase and/or magnitude based on a corresponding control signal **S** received from control circuitry **38** of FIG. **4** over control paths **34** (e.g., the phase and/or magnitude provided by phase and magnitude controller **33-1** may be controlled using control signal **S1** on control path **34-1**, the phase and/or magnitude provided by phase and magnitude controller **33-2** may be controlled using control signal **S2** on control path **34-2**, the phase and/or magnitude provided by phase and magnitude controller **33-M** may be controlled using control signal **SM** on control path **34-M**, etc.). If desired, control circuitry **38** may actively adjust control signals **S** in real time to steer the transmit or receive beam in different desired directions (e.g., to different desired beam pointing angles) over time. Phase and magnitude controllers **33** may provide information identifying the phase of received signals to control circuitry **38** if desired.

[0054] When performing wireless communications using radio-frequency signals at millimeter and centimeter wave frequencies, the radio-frequency signals are conveyed over a line of sight path between phased antenna array **36** and external wireless equipment (e.g., external wireless equipment **10'** of FIG. **3**). If the external wireless equipment is located at point A of FIG. **4**, phase and magnitude controllers **33** may be adjusted to steer the signal beam towards point A (e.g., to form a signal beam having a beam pointing angle directed towards point A). Phased antenna array **36** may then transmit and receive radio-frequency signals in the direction of point A. Similarly, if the external wireless equipment is located at point B, phase and magnitude controllers **33** may be adjusted to steer the signal beam towards point B (e.g., to form a signal beam having a beam pointing angle directed towards point B). Phased antenna array **36** may then transmit and receive radio-frequency signals in the direction of point B. In the example of FIG. **4**, beam steering is shown as being performed over a single degree of freedom for the sake of simplicity (e.g., towards the left and right on the page of FIG. **4**). However, in practice, the beam may be steered over two or more degrees of freedom (e.g., in three dimensions, into and out of the page and to the left and right on the page of FIG. **4**). Phased antenna array **36** may have a corresponding field of view over which beam steering can be performed (e.g., in a hemisphere or a segment of a hemisphere over the phased antenna array). If desired, device **10** may include multiple phased antenna arrays that each

face a different direction to provide coverage from multiple sides of the device.

[0055] Control circuitry **38** of FIG. **4** may form a part of control circuitry **14** of FIG. **3** or may be separate from control circuitry **14** of FIG. **3**. Control circuitry **38** of FIG. **4** may identify a desired beam pointing angle for the signal beam of phased antenna array **36** and may adjust the control signals **S** provided to phased antenna array **36** to configure phased antenna array **36** to form (steer) the signal beam at that beam pointing angle. Each possible beam pointing angle that can be used by phased antenna array **36** during wireless communications may be identified by a beam steering codebook such as codebook **40**. Codebook **40** may be stored at control circuitry **38**, elsewhere on device **10**, or may be located (offloaded) on external equipment and conveyed to device **10** over a wired or wireless communications link.

[0056] Codebook **40** may identify each possible beam pointing angle that may be used by phased antenna array **36**. Control circuitry **38** may store or identify phase and magnitude settings for phase and magnitude controllers **33** to use in implementing each of those beam pointing angles (e.g., control circuitry **38** or codebook **40** may include information that maps each beam pointing angle for phased antenna array **36** to a corresponding set of phase and magnitude values for phase and magnitude controllers **33**). Codebook **40** may be hard-coded or soft-coded into control circuitry **38** or elsewhere in device **10**, may include one or more databases stored at control circuitry **38** or elsewhere in device **10** (e.g., codebook **40** may be stored as software code), may include one or more look-up-tables at control circuitry **38** or elsewhere in device **10**, and/or may include any other desired data structures stored in hardware and/or software on device **10**. Codebook **40** may be generated during calibration of device **10** (e.g., during design, manufacturing, and/or testing of device **10** prior to device **10** being received by an end user) and/or may be dynamically updated over time (e.g., after device **10** has been used by an end user).

[0057] Control circuitry **38** may generate control signals **S** based on codebook **40**. For example, control circuitry **38** may identify a beam pointing angle that would be needed to communicate with external wireless equipment **10'** of FIG. **3** (e.g., a beam pointing angle pointing towards external wireless equipment **10'**). Control circuitry **38** may subsequently identify the beam pointing angle in codebook **40** that is closest to this identified beam pointing angle. Control circuitry **38** may use codebook **40** to generate phase and magnitude values for phase and magnitude controllers **33**. Control circuitry **38** may transmit control signals **S** identifying these phase and magnitude values to phase and magnitude controllers **33** over control paths **34**. The beam formed by phased antenna array **36** using control signals **S** will be oriented at the beam pointing angle identified by codebook **40**. If desired, control circuitry **38** may sweep over some or all of the different beam pointing angles identified by codebook **40** until the external wireless equipment is found and may use the corresponding beam pointing angle at which the external wireless equipment was found to communicate with the external wireless equipment (e.g., over communications link **31** of FIG. **3**).

[0058] A schematic diagram of an antenna **30** that may be formed in phased antenna array **36** (e.g., as antenna **30-1**, **30-2**, **30-3**, and/or **30-N** in phased antenna array **36** of FIG. **4**) is shown in FIG. **5**. As shown in FIG. **5**, antenna **30** may be coupled to transceiver circuitry **42** (e.g., millimeter wave transceiver circuitry **28** of FIG. **3**). Transceiver circuitry **42** may be coupled to antenna feed **48** of antenna **30** using radio-frequency transmission line path **32**. Antenna feed **48** may include a positive antenna feed terminal such as positive antenna feed terminal **50** and may include a ground antenna feed terminal such as ground antenna feed terminal **52**. Radio-frequency transmission line path **32** may include a positive signal conductor such as signal conductor **44** that is coupled to positive antenna feed terminal **50** and a ground conductor such as ground conductor **46** that is coupled to ground antenna feed terminal **52**.

[0059] Any desired antenna structures may be used for implementing antenna **30**. In one suitable arrangement that is sometimes described herein as an example, stacked patch antenna structures may be used for implementing antenna **30**. Antennas **30** that are implemented using stacked patch antenna structures may sometimes be referred to herein as stacked patch antennas or simply as

patch antennas. FIG. 6 is a top view of an illustrative patch antenna that may be used in phased antenna array 36.

[0060] As shown in FIG. 6, antenna 30 may have an antenna radiating element that includes patch element 58. Patch element 58 (sometimes referred to herein as patch 58 or conductive patch 58) may be formed from conductive traces on an underlying substrate or from any other desired conductive materials. Patch element 58 may be separated from and extend parallel to an antenna ground (not shown in FIG. 6 for the sake of clarity).

[0061] Patch element 58 may have edges (sides) 66. The length of edges 66 may be selected so that antenna 30 resonates (radiates) at desired operating frequencies. In one suitable arrangement that is described herein as an example, patch element 58 is a square patch having edges 66 of length L1 (e.g., where patch element 58 has a first pair of parallel edges 66 and a second pair of parallel edges 66 extending orthogonal to and between the first pair of parallel edges 66). Length L1 may be selected to be approximately equal to half of the wavelength of the signals conveyed by antenna 30 (e.g., the effective wavelength given the dielectric properties of the materials surrounding patch element 58). In one suitable arrangement, this length may be between 0.8 mm and 1.2 mm (e.g., approximately 1.1 mm) for covering a millimeter wave frequency band between 57 GHz and 70 GHz, as just one example. The example of FIG. 6 merely illustrative. If desired, patch element 58 may have a non-square rectangular shape having two edges of length L1 and having two edges of a different length (e.g., for covering multiple frequency bands). In general, patch element 58 may be formed in any desired shape having any desired number of straight and/or curved edges.

[0062] To enhance the polarizations handled by antenna 30, antenna 30 may be provided with multiple antenna feeds. As shown in FIG. 6, antenna 30 may include a first antenna feed having positive antenna feed terminal 50A and may include a second antenna feed having positive antenna feed terminal 50B. Positive antenna feed terminals 50A and 50B may be coupled to transceiver circuitry 42 (FIG. 5) using respective radio-frequency transmission line paths 32, for example. Positive antenna feed terminals 50A and 50B may be coupled to patch element 58.

[0063] When using positive antenna feed terminal 50A, antenna 30 may transmit and/or receive radio-frequency signals with a first polarization (e.g., a first linear polarization). When using positive antenna feed terminal 50B, antenna 30 may transmit and/or receive radio-frequency signals with a second polarization (e.g., a second linear polarization). The second polarization may be orthogonal to the first polarization. This is merely illustrative and, if desired, positive antenna feed terminals 50A and 50B may be used to convey radio-frequency signals with other polarizations (e.g., elliptical polarizations, circular polarizations, etc.). Antenna 30 may include only one of positive antenna feed terminals 50A or 50B if desired (e.g., antenna 30 need not be a dual-polarization antenna).

[0064] In order to increase the bandwidth of antenna 30, antenna 30 may include one or more parasitic elements layered over (e.g., overlapping) patch element 58. As shown in FIG. 6, a parasitic antenna resonating element such as parasitic patch 68 may be formed from conductive traces layered over patch element 58. Patch element 58 may, for example, be formed from conductive traces patterned onto a first layer of a dielectric substrate whereas parasitic patch 68 is formed from conductive traces patterned onto a second layer of the dielectric substrate (e.g., where the first and second layers are vertically stacked on top of each other in the direction of the Z-axis of FIG. 6).

[0065] Parasitic patch 68 may sometimes be referred to herein as parasitic resonating element 68, parasitic antenna element 68, parasitic element 68, parasitic conductor 68, parasitic structure 68, or patch 68. Parasitic patch 68 is not directly fed, whereas patch element 58 is directly fed via positive antenna feed terminals 50A and 50B. Parasitic patch 68 may create a constructive perturbation of the electromagnetic field generated by patch element 58, creating a new resonance for antenna 30. This may serve to broaden the overall bandwidth of antenna 30 (e.g., to cover an entire frequency band from about 57 GHz to 71 GHz).

[0066] In one suitable arrangement that is described herein as an example, parasitic patch **68** is a square patch having edges (sides) of length **L2**. The edges of parasitic patch **68** may be oriented parallel to the edges **66** of patch element **58** (e.g., parasitic patch **68** may be aligned with patch element **58**). Length **L2** may be less than length **L1** of patch element **58**. The example of FIG. **6** merely illustrative. If desired, parasitic patch **68** may have a non-square rectangular shape or any other desired shape having any desired number of straight and/or curved edges. If desired, antenna **30** may include additional parasitic elements that are coplanar with parasitic patch **68**.

[0067] For example, as shown in FIG. **6**, antenna **30** may include additional parasitic patches **64** (sometimes be referred to herein as parasitic resonating elements **64**, parasitic antenna elements **64**, parasitic elements **64**, parasitic conductors **64**, parasitic structures **64**, or patches **64**). Parasitic patches **64** may be coplanar with parasitic patch **102**. Each parasitic patch **64** may be separated from a corresponding edge of parasitic patch **68** by a respective gap **62**. Each parasitic patch **64** may, if desired, overlap a respective edge **66** of the underlying patch element **58**. Each parasitic patch **64** may be the same size and shape, for example.

[0068] In one suitable arrangement that is described herein as an example, parasitic patches **64** are rectangular patches having edges (sides) that are shorter than length **L1** and that are greater than, equal to, or less than length **L2**. The example of FIG. **6** merely illustrative. If desired, parasitic patches **64** may have other non-square rectangular shapes or any other desired shapes having any desired number of straight and/or curved edges. Gaps **62** (sometimes referred to herein as openings **62** or slots **62**) may help to mitigate the trapping of radio-frequency energy between the parasitic elements and patch element **58**, for example. Parasitic patches **64** and **68** may sometimes be referred to herein collectively as single-layer parasitic antenna resonating elements, single-layer parasitic elements, single-layer parasitic patches, or single-layer parasitic structures for antenna **30**.

[0069] Each antenna **30** in phased antenna array **36** may include a corresponding patch element **58** and overlying single-layer parasitic structures. The antennas **30** in phased antenna array **36** may be arranged in an array pattern having any desired number of rows (e.g., extending along a longitudinal axis parallel to the X-axis) and/or any desired number of columns (e.g., extending along a longitudinal axis parallel to the Y-axis). If care is not taken, the antennas **30** in phased antenna array **36** may exhibit undesirable mutual coupling with one or more adjacent antennas **30** in phased antenna array **36**. Such mutual coupling can undesirably limit the overall antenna efficiency of each antenna **30**. In order to mitigate mutual coupling in phased antenna array **36**, antenna **30** may be diagonally-oriented with respect to the rows and columns of phased antenna array **36** and may include diagonally-oriented positive antenna feed terminals **50A** and **50B**.

[0070] For example, as shown in FIG. **6**, patch element **58**, parasitic patch **68**, and parasitic patches **64** may be rotated (e.g., about a central axis **60** extending parallel to the Z-axis) at a non-zero angle with respect to the direction of the rows and columns in phased antenna array **36** (e.g., with respect to the X and Y-axes of FIG. **6**). In one suitable arrangement that is described herein as an example, the non-zero angle is 45 degrees. Other non-zero angles may be used if desired (e.g., 40-50 degrees, 35-55 degrees, 44-46 degrees, etc.).

[0071] Patch element **58** may have a first diagonal axis **54** and a second diagonal axis **56**. Diagonal axis **54** may extend through central axis **60** and a first pair of opposing corners of patch element **58**. Diagonal axis **56** may be perpendicular to diagonal axis **54**. Diagonal axis **56** may extend through central axis **60** and a second pair of opposing corners of patch element **58**. As parasitic patch **68** is also centered about central axis **60**, diagonal axis **56** also passes through a first pair of opposing corners of parasitic patch **68**. Similarly, diagonal axis **54** also passes through a second pair of opposing corners of parasitic patch **68**.

[0072] When oriented in this way, each of the antennas **30** along a given row of phased antenna array **36** may have a central axis (e.g., central axis **60**) that intersects the diagonal axis **54** of each antenna **30** in that row of phased antenna array **36**. Similarly, each of the antennas **30** along a given column of phased antenna array **36** may have a central axis that intersects the diagonal axis **56** of

each antenna **30** in that column of phased antenna array **36**. In other words, diagonal axis **54** may form the longitudinal axis for a given row of antennas **30** (e.g., where each antenna **30** in the row is aligned along the longitudinal axis for that row) and diagonal axis **56** may form the longitudinal axis for a given column of antennas **30** in phased antenna array **36** (e.g., where each antenna **30** in the column is aligned along the longitudinal axis for that column). When oriented in this way, edges **66** of patch element **58** and the edges of parasitic patch **68** are each oriented at the non-zero angle (e.g., 45 degrees) with respect to diagonal axes **56** and **54** and with respect to the direction (e.g., the longitudinal axes) of the rows and the columns in phased antenna array **36**.

[0073] Diagonally orienting the antennas **30** in phased antenna array **36** in this way may serve to minimize mutual coupling between the antennas in the phased antenna array, thereby maximizing the overall antenna efficiency of each of the antennas. In order to further mitigate mutual coupling and optimize antenna efficiency (e.g., relative to scenarios where positive antenna feed terminals **50A** and **50B** are located along respective edges **66** of patch element **58**), positive antenna feed terminal **50A** may be coupled to patch element **58** at a location along diagonal axis **56**. Similarly, positive antenna feed terminal **50B** may be coupled to patch element **58** at a location along diagonal axis **54**. The distance between positive antenna feed terminal **50A** and central axis **60** (e.g., along diagonal axis **56**) and the distance between positive antenna feed terminal **50B** and central axis **60** (e.g., along diagonal axis **54**) may be selected to perform impedance matching for antenna **30**, for example. Feeding antenna **30** in this way may also allow antenna **30** to continue to convey linearly-polarized signals (e.g., horizontal and vertically polarized signals) using positive antenna feed terminals **50A** and **50B**, for example.

[0074] FIG. 7 is a cross-sectional side view of antenna **30** (e.g., as taken in the direction of line AA' of FIG. 6). As shown in FIG. 7, antenna **30** may be formed on a dielectric substrate such as substrate **70**. If desired, each of the antennas in the phased antenna array may be formed on the same dielectric substrate (e.g., in an integrated antenna module having a radio-frequency integrated circuit mounted to substrate **70**). Substrate **70** may be, for example, a rigid or printed circuit board or another dielectric substrate. Substrate **70** may include multiple stacked dielectric layers **72** (e.g., layers of printed circuit board substrate, layers of fiberglass-filled epoxy, layers of polyimide, layers of ceramic substrate, or layers of other dielectric materials).

[0075] With this type of arrangement, antenna **30** may be embedded within the layers of substrate **70**. For example, patch element **58** may be formed from conductive traces **92** patterned on a first layer **72** of substrate **70**. Parasitic patches **68** and **64** may be formed from conductive traces **90** patterned on a second layer **72** of substrate **70**. The second layer may be stacked over the first layer of substrate **70**. Zero, one, or more than one additional layer **72** may be vertically interposed between the first and second layers **72** if desired. Gaps **62** in conductive traces **90** may separate parasitic patch **68** from parasitic patches **64**.

[0076] Antenna **30** may have an antenna ground that includes ground traces **74** (e.g., a ground plane for antenna **30**). The same ground traces **74** may be used to form the antenna ground for each antenna in the phased antenna array if desired. Patch element **92** may be separated from and may extend parallel to ground traces **74**. One or more layers **72** of substrate **70** may be vertically interposed between ground traces **74** and patch element **58**. Zero, one, or more than one layer **72** in substrate **70** may be vertically interposed between conductive traces **90** and the exterior of substrate **70**.

[0077] Ground traces **74** may have openings such as opening **76**. Signal traces **80** may be patterned on one or more of the layers **72** in substrate **70** (e.g., ground traces **74** may be vertically interposed between signal traces **80** and patch element **58**). Signal traces **80** may, for example, form the signal conductor of the radio-frequency transmission line path for antenna **30** (e.g., signal conductor **44** in radio-frequency transmission line path **32** of FIG. 5). A conductive via such as conductive via **78** may couple signal traces **80** to patch element **58** (e.g., at positive antenna feed terminal **50B**). Similar feeding structures may be used to feed positive antenna feed terminal **50A** (FIG. 6). As

shown in FIG. 7, parasitic patches **68** and **64** are not directly fed by positive antenna feed terminal **50B**.

[0078] FIG. **8** is a plot of antenna performance (mutual coupling) as a function of frequency for a given antenna **30** in phased antenna array **36** (FIG. **6**). As shown in FIG. **8**, curve **94** plots the mutual coupling of antenna **30** in scenarios where the antennas are not rotated by the non-zero angle with respect to the X and Y axes of FIG. **6** and where the antennas are fed using positive antenna feed terminals **50A** and **50B** located along orthogonal edges **66** of patch element **58**.

[0079] Curve **98** plots the mutual coupling of antenna **30** in scenarios where the antennas in the phased antenna array are oriented and fed as shown in FIG. **6**. As shown by curves **98** and **94**, rotating the antenna elements and feeding the antenna along diagonal axes **54** and **56** may serve to reduce mutual coupling across the frequency band of operation of antenna **30**, as shown by arrow **96**. This reduction in mutual coupling may serve to increase the overall antenna efficiency of antenna **30**, for example. The example of FIG. **8** is merely illustrative. In practice, curves **94** and **98** may have other shapes. Antenna **30** may convey radio-frequency signals at any desired frequencies (e.g., frequencies greater than 10 GHz).

[0080] In the example of FIGS. **6-8**, the parasitic patches in antenna **30** are confined to a single layer **72** of substrate **70**. If desired, the parasitic patches in antenna **30** may be distributed across two or more layers **72** of substrate **70**. FIG. **9** is a top view of an antenna **30** having parasitic patches distributed across multiple layers of the substrate.

[0081] As shown in FIG. **9**, antenna **30** may include a parasitic patch such as parasitic patch **102** (sometimes referred to herein as parasitic resonating element **102**, parasitic antenna element **102**, parasitic element **102**, parasitic conductor **102**, parasitic structure **102**, or patch **102**). Parasitic patch **102** and patch element **58** may be centered about central axis **60**. In one suitable arrangement that is described herein as an example, parasitic patch **102** is a square patch having edges (sides) of length **L3**. Length **L3** may be less than the length of the edges of patch element **58** (e.g., length **L1** as shown in FIG. **6**). The edges of parasitic patch **102** may be oriented parallel to the edges of patch element **58** (e.g., parasitic patch **102** may be aligned with patch element **58**). The example of FIG. **9** merely illustrative. If desired, parasitic patch **102** may have a non-square rectangular shape or any other desired shape having any desired number of straight and/or curved edges.

[0082] Antenna **30** may also include additional parasitic patches **104** (sometimes be referred to herein as parasitic resonating elements **104**, parasitic antenna elements **104**, parasitic elements **104**, parasitic conductors **104**, parasitic structures **104**, or patches **104**). Parasitic patches **104** may be located at a different distance from patch element **58** than parasitic patch **102**. For example, parasitic patches **104** may be located at a first distance from (over) patch element **58** whereas parasitic patch **102** is located at a second distance that is greater than the first distance from patch element **58**. Each parasitic patch **104** may be separated from an opposing parasitic patch **104** by gap **100**. Gap **100** may overlap patch element **58** and central axis **60**. Parasitic patch **102** may overlap gap **100**. In the example of FIG. **9**, parasitic patch **102** is non-overlapping with respect to parasitic patches **104**. In another suitable arrangement, parasitic patches **104** may partially overlap parasitic patch **102**. Each parasitic patch **104** may, if desired, overlap a respective edge of the underlying patch element **58**.

[0083] If desired, each parasitic patch **104** may be the same size and shape. In one suitable arrangement that is described herein as an example, parasitic patches **104** are rectangular patches having edges (sides) that are shorter than length **L1** (FIG. **6**) and that are greater than, equal to, or less than length **L3**. Each parasitic patch **104** may have edges that are oriented parallel to the edges of patch element **58** and parasitic patch **102**. The example of FIG. **9** merely illustrative. If desired, parasitic patches **104** may have other rectangular shapes or any other desired shapes having any desired number of straight and/or curved edges. Parasitic patches **104** and **102** may sometimes be referred to herein collectively as multi-layer parasitic antenna resonating elements, multi-layer parasitic elements, multi-layer parasitic patches, or multi-layer parasitic structures for antenna **30**.

[0084] In the example of FIG. 9, the edges of parasitic patches **104** and **102** and the edges of patch element **58** are oriented parallel to the direction of the rows and columns in phased antenna array **36**. Positive antenna feed terminals **50A** and **50B** may be coupled to patch element **58** along orthogonal edges of patch element **58**. This example is merely illustrative. In another suitable arrangement, parasitic patches **104** and **102** and patch element **58** may be rotated at a non-zero (e.g., 45 degree) angle with respect to the direction of the rows and columns in phased antenna array **36** and patch element **58** may be fed along the diagonal axes of patch element **58** (e.g., antenna **30** may be rotated and fed as shown in FIG. 6 but may include the multi-layer parasitic structures of FIG. 9).

[0085] FIG. 10 is a cross-sectional side view of antenna **30** having multi-layer parasitic structures (e.g., as taken in the direction of line BB' of FIG. 9). As shown in FIG. 10, patch element **58** may be formed from conductive traces **92** patterned on a first layer **72** of substrate **70**. Parasitic patches **104** may be formed from conductive traces **108** patterned on a second layer **72** of substrate **70**. The second layer may be stacked over the first layer of substrate **70**. Zero, one, or more than one additional layer **72** may be vertically interposed between the first and second layers **72** if desired. Parasitic patch **102** may be formed from conductive traces **106** patterned on a third layer **72** of substrate **70**. The third layer may be stacked over the second layer of substrate **70**. Zero, one, or more than one additional layer **72** may be vertically interposed between the second and third layers **72** if desired.

[0086] Parasitic patches **104** may be separated by gap **100** overlapping patch element **58**. Parasitic patch **102** may overlap gap **100** and patch element **58**. Patch element **58** may be directly fed whereas parasitic patches **104** and **102** are not directly fed (e.g., each of the parasitic patches is floating). First capacitances may be established between parasitic patch **102** and each of the parasitic patches **104**. Second capacitances may be established between each of the parasitic patches **104** and patch element **58**. These capacitances may serve to increase the total capacitance between patch element **58** and the upper-most parasitic patch relative to arrangements where antenna **30** includes single-layer parasitic structures, which may allow antenna **30** to exhibit an even more compact volume relative to arrangements where antenna **30** includes single-layer parasitic structures, for example.

[0087] When arranged in this way, the parasitic patches may provide freedom to fine tune the radio-frequency performance of antenna **30** for compensating for changes in dielectric thickness, dielectric constant, radome material (e.g., for a radome placed over antenna **30**), copper thickness, etc., without changing the antenna radiation mechanism or radiation pattern. In other words, the lateral footprint of antenna **30** of FIGS. 9 and 10 (e.g., as defined by a square running through the outer-most edges of parasitic patches **104** as shown in FIG. 9) may be smaller than the lateral footprint of antenna **30** of FIGS. 6 and 7 (e.g., as defined by a rotated square running through the outer-most edges of parasitic patches **64** as shown in FIG. 6). Conversely, when antenna **30** of FIGS. 9 and 10 is configured to exhibit the same lateral footprint as antenna **30** of FIGS. 6 and 7, antenna **30** may exhibit increased bandwidth relative to antenna **30** of FIGS. 6 and 7.

[0088] FIG. 11 is a plot of antenna performance (return loss) as a function of frequency for a given antenna **30** (e.g., an antenna **30** having a given lateral footprint). As shown in FIG. 11, curve **110** plots the return loss of an antenna **30** having single-layer parasitic structures (e.g., antenna **30** of FIGS. 6 and 7). Curve **112** plots the return loss of an antenna **30** having multi-layer parasitic structures (e.g., antenna **30** of FIGS. 9 and 10). As shown by curves **110** and **112**, antenna **30** may exhibit satisfactory return loss (e.g., a return loss less than threshold level TH) across the frequency band of operation of the antenna. However, forming antenna **30** using multi-layer parasitic structures (e.g., as shown in FIGS. 9 and 10) may further reduce the return loss of the antenna, as shown by arrow **114**.

[0089] The example of FIG. 11 is merely illustrative. In practice, curves **110** and **112** may have other shapes. Antenna **30** may convey radio-frequency signals at any desired frequencies (e.g.,

frequencies greater than 10 GHz).

[0090] Device **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

[0091] The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. Apparatus comprising: a substrate; and an antenna on the substrate that includes a first patch, wherein the first patch has a first corner, a second corner, and a diagonal axis passing through the first and second corners, a conductive via in the substrate and coupled to the first patch along the diagonal axis, a second patch overlapping the first patch, a third patch separated from a first edge of the second patch by a first gap, a fourth patch separated from a second edge of the second patch by a second gap, a fifth patch separated from a third edge of the second patch by a third gap, and a sixth patch separated from a fourth edge of the second patch by a fourth gap.
2. The apparatus of claim 1, wherein the first patch is directly fed.
3. The apparatus of claim 2, wherein the second patch is parasitic.
4. The apparatus of claim 3, wherein the third, fourth, fifth, and sixth patches are parasitic.
5. The apparatus of claim 1, wherein the second patch has a first length, the third patch has a second length equal to the first length, the fourth patch has a third length equal to the first length, the fifth patch has a fourth length equal to the first length, and the sixth patch has a fifth length equal to the first length.
6. The apparatus of claim 1, wherein the third patch at least partially overlaps the first patch, the fourth patch at least partially overlaps the first patch, the fifth patch at least partially overlaps the first patch, and the sixth patch at least partially overlaps the first patch.
7. The apparatus of claim 6, wherein the first patch has a fifth edge parallel to the first edge, a sixth edge parallel to the second edge, a seventh edge parallel to the third edge, and an eighth edge parallel to the fourth edge.
8. The apparatus of claim 7, wherein the third patch overlaps the fifth edge, the fourth patch overlaps the sixth edge, the fifth patch overlaps the seventh edge, and the sixth patch overlaps the eighth edge.
9. The apparatus of claim 1, wherein the first and second edges of the second patch extend parallel to each other and have a first length, the third and fourth edges of the second patch extend orthogonal to the first and second edges and have the first length, each of the third, fourth, fifth, and sixth patches has a second length equal to the first length, and each of the third, fourth, fifth, and sixth patches has a width less than the second length.
10. The apparatus of claim 1, further comprising: a transmission line having a signal trace coupled to the conductive via.
11. The apparatus of claim 1, wherein the first patch has a third and fourth corners and an additional diagonal axis passing through the third corner and the fourth corner, the apparatus comprising: an additional conductive via in the substrate and coupled to the first patch along the additional diagonal axis.
12. The apparatus of claim 11, further comprising: an additional antenna on the substrate, wherein the additional antenna includes a seventh patch embedded in the substrate, the seventh patch is rectangular, has a fifth corner, a sixth corner opposite the fifth corner, a seventh corner, and an

eighth corner opposite the seventh corner, and the diagonal axis of the first patch passes through the fifth corner and the sixth corner of the additional directly fed patch.

13. Apparatus comprising: a substrate; and a phased antenna array that includes a first antenna and a second antenna on the substrate, wherein the first antenna includes a first patch fed by a first conductive via in the substrate, a second patch overlapping the first patch, a third patch separated from a first edge of the second patch by a first gap, a fourth patch separated from a second edge of the second patch by a second gap, a fifth patch separated from a third edge of the second patch by a third gap, and a sixth patch separated from a fourth edge of the second patch by a fourth gap, the second antenna including a seventh patch fed by a second conductive via in the substrate, a center of the first patch and a center of the seventh patch being aligned along a linear axis, the seventh patch being square, the first, second, third, and fourth edges of the first patch being non-parallel and non-perpendicular with respect to the linear axis, and four edges of the seventh patch being non-parallel and non-perpendicular with respect to the linear axis.

14. The apparatus of claim 13, wherein the second antenna further comprises: an eighth patch overlapping the seventh patch, a ninth patch separated from a fifth edge of the eighth patch by a fifth gap, a tenth patch separated from a sixth edge of the eighth patch by a sixth gap, an eleventh patch separated from a seventh edge of the eighth patch by a seventh gap, and a twelfth patch separated from an eighth edge of the eighth patch by an eighth gap.

15. The apparatus of claim 14, wherein: the eighth patch is square, the ninth, tenth, eleventh, and twelfth patches are polygonal and non-square, the linear axis overlaps first and second corners of the second patch, the linear axis overlaps third and fourth corners of the eighth patch, and the linear axis does not overlap the third, fourth, fifth, sixth, ninth, tenth, eleventh, and twelfth patches.

16. The apparatus of claim 13, wherein the first edge is parallel to the third edge, the first gap extends parallel to the third gap, the second edge is parallel to the fourth edge, the second gap extends parallel to the fourth gap, the first edge is orthogonal to the second edge, and the second gap extends orthogonal to the first gap.

17. The electronic device of claim 13, wherein: the first patch is square, the first patch is fed at a first positive antenna feed terminal coupled to the first patch along a diagonal axis of the first patch, the seventh patch is square, and the seventh patch is fed at a second positive antenna feed terminal coupled to the seventh patch along a diagonal axis of the seventh patch.

18. An antenna module comprising: a substrate; and an antenna on the substrate, wherein the antenna includes a rectangular metal patch having a diagonal axis, an antenna feed terminal coupled to the rectangular metal patch along the diagonal axis, a first patch overlapping the rectangular metal patch, a second patch separated from a first edge of the first patch by a first gap, a third patch separated from a second edge of the first parasitic patch by a second gap, a fourth patch separated from a third edge of the first parasitic patch by a third gap, and a fifth patch separated from a fourth edge of the first parasitic patch by a fourth gap.

19. The antenna module of claim 18, wherein the substrate comprises a printed circuit board substrate, the second, third, fourth, and fifth patches being parasitic.

20. The antenna module of claim 18, further comprising a conductive via in the substrate and coupled to the rectangular metal patch at the antenna feed terminal.
